

Mathematical Analysis Zorich Comprehensions

Buce-Ithon

13th August, 2025

Contents

1	Multiple Integrals	4
1.1	The Riemann Integral over an n -Dimensional Interval	4
1.1.1	Comprehensions	4
1.1.2	Problems Sets	5



Figure 1: One week, one Zorich

1 Multiple Integrals

1.1 The Riemann Integral over an n -Dimensional Interval

1.1.1 Comprehensions

First, the definition of Riemann integral over n -dimension interval is simimiar (at most same) with 1-dimension condition ($\int_I f(x)dx := \lim_{\lambda(P) \rightarrow 0} \sigma(f, P, \xi)$),

$$\int_I f(x^1, \dots, x^n) dx^1 \dots dx^n \text{ or } \int \dots \int_I f(x^1, \dots, x^n) dx^1 \dots dx^n$$

Then, introduce the nessasary condition of it ($f \in \mathcal{R}(I) \implies f$ is bounded on I). After that, book gives some properties (Lemmas) about the set of measure zero (same with 1-dimension condition), and developes the Cantor Theorem (about uniform continuity of continuous functions on compact set). At last, we're given 2 criterions (nessasary and sufficient) of the Riemann integral (Darboux Criterion, Lebesgue Criterion).

Example. Let $f : I \rightarrow \mathbb{R}$ is a continuous real-value function on $I \subset \mathbb{R}^{n-1}$, then it's graph is a set measure zero in n -dimension.

Lemma 1.1. /Development of Cantor Theorem/ If the relation $\omega(f; x) < w_0$ holds at each point of a compact set K for the function $f : K \rightarrow \mathbb{R}$, then for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\omega(f; U_\delta(x)) < w_0 + \varepsilon$ for each point $x \in K$.

Theorem 1.1. (Lebesgue's criterion). $f \in \mathcal{R}(I) = (f \text{ is bounded on } I) \wedge (f \text{ is continuous almost everywhere on } I)$.

Theorem 1.2. (Darboux Criterion) $f \in (I) \Leftrightarrow (f \text{ is bounded on } I) \wedge (\underline{\mathcal{J}} = \overline{\mathcal{J}})$, where $\underline{\mathcal{J}} = \sup_P s(f, P)$, $\overline{\mathcal{J}} = \inf_P S(f, P)$.

1.1.2 Problems Sets

Problem 1. a) Show that a set of measure zero has no interior points.

- b) Show that not having interior points by no means guarantees that a set is of measure zero.
- c) Construct a set having measure zero whose closure is the entire space $\mathbb{R}^n$.
- d) A set $E \subset I$ is said to have content zero if for every $\varepsilon > 0$ it can be covered by a finite system of intervals $I_1, \dots, I_k$ such that $\sum_{i=1}^k |I_i| < \varepsilon$. Is every bounded set of measure zero a set of content zero?
- e) Show that if a set $E \subset \mathbb{R}^n$ is the direct product $\mathbb{R} \times e$ of the line $\mathbb{R}$ and a set $e \subset \mathbb{R}^{n-1}$ of $(n-1)$ -dimensional measure zero, then E is a set of n -dimensional measure zero.

Proof. a) [Proof by contradiction] If E has a point x_0 as an interior point, then there exists a ball $B(x_0, r)$ such that $B(x_0, r) \subset E$. Since the ball has positive measure, it follows that E cannot have measure zero.

b) [Counterexample] Consider the irrational numbers $\mathbb{R} \setminus \mathbb{Q}$, which have no interior points in $\mathbb{R}$ but are not of measure zero.

c) /inspired by b)/ Consider the set $E = \mathbb{Q}^n$, the set of all points in $\mathbb{R}^n$ with rational coordinates. The closure of E is $\mathbb{R}^n$, and E has measure zero since it can be covered by countably many intervals whose total length can be made arbitrarily small.

d) [Principle] No, since the Lebesgue measure zero set can be covered by at most countable intervals (which means measure $\mu = o(1) \cdot \mathcal{O}(1)$), while a set of content zero can be covered by a finite number of intervals (which means the content (also a kind of measure) sometimes is $\mathcal{O}(1) \cdot \mathcal{O}(1)$). [Example] The $\mathbb{Q} \cap [0, 1]$ and Cantor set are sets of measure zero but none of content zero.

e) From the construction of the product measure λ_n (Lebesgue measure of $\mathbb{R}^n$) /Lemma1/ if $A \in \mathcal{B}(\mathbb{R}^k)$ and $B \in \mathcal{B}(\mathbb{R}^{n-k})$, ($0 < k < n$), then $\lambda_n(A \times B) = \lambda_k(A)\lambda_{n-k}(B)$. So here (using also that /Lemma2/ if $E_m \uparrow E$, $\lim_{m \rightarrow +\infty} \lambda_n(E_m) = \lambda_n(E)$) $\lambda_n(E) = \lambda_n(\mathbb{R} \times e) = \lim_{m \rightarrow +\infty} \lambda_n((-m, m) \times e) = \lim_{m \rightarrow +\infty} \lambda_1((-m, m)) \times \lambda_{n-1}(e) = \lim_{m \rightarrow +\infty} \lambda_1((-m, m)) \times 0 = 0$.

Notes. 1. The proof of e) /Lemma2/ could found in *Real and Complex Analysis* by Walter Rudin, Theorem 1.19(Page 16), based on the (positive) measure definition (countable additive of disjoint countable measurable sets).

- Problem 2.** a) Construct the analogue of the Dirichlet function in $\mathbb{R}^n$ and show that a bounded function $f : I \rightarrow \mathbb{R}$ equal to zero at almost every point of the interval I may still fail to belong to $\mathcal{R}(I)$.
- b) Show that if $f \in \mathcal{R}(I)$ and $f(x) = 0$, at almost all points of the interval I , then $\int_I f(x) dx = 0$.

Proof. a) [Counterexample] Consider $\mathcal{D}^n = \begin{cases} 1 & \text{if } x \in \mathbb{Q}^n \\ 0 & \text{if } x \in \mathbb{R}^n \setminus \mathbb{Q}^n \end{cases}$, the discontinuous set is $\mathcal{R}^n$.

b) /by definition/ $\int_I f(x) dx = \lim_{\lambda(P) \rightarrow 0} \sum_{\xi_i \in P} f(\xi_i) \Delta(x_i) = \lim_{\lambda(P) \rightarrow 0} \sum_{\xi_i \in A \cap P} f(\xi_i) \Delta(x_i) + \lim_{\lambda(P) \rightarrow 0} \sum_{\xi_j \in B \cap P} f(\xi_j) \Delta(x_i) = \mathcal{O}(1) \cdot o(1) + o(1) \cdot \mathcal{O}(1)$, where $A = \{\cup [x_k, x_{k+1}] \subset P : \forall k, \exists \xi \in [x_k, x_{k+1}], \text{ s.t. } f(\xi) \neq 0\}$ and $B = I \setminus A$.

Problem 3. There is a small difference between our earlier definition of the Riemann integral on a closed interval $I \subset \mathbb{R}$ and Definition 7 for the integral over an interval of arbitrary dimension. This difference involves the definition of a partition and the measure of an interval of the partition. Clarify this nuance for yourself and verify that

$$\int_a^b f(x) dx = \begin{cases} \int_I f(x) dx & \text{if } a < b \\ - \int_I f(x) dx & \text{if } a > b \end{cases}$$

where I is the interval on the real line $\mathbb{R}$ with endpoints a and b .

Proof. Just understand it, it's not hard.

- Problem 4.** a) Prove that a real-valued function $f : I \rightarrow \mathbb{R}$ defined on an interval $I \subset \mathbb{R}^n$ is integrable over that interval if and only if for every $\varepsilon > 0$ there exists a partition P of I such that $S(f; P) - s(f; P) < \varepsilon$.
- b) Using the result of a) and assuming that we are dealing with a real-valued function $f : I \rightarrow \mathbb{R}$, one can simplify slightly the proof of the sufficiency of the Lebesgue criterion. Try to carry out this simplification by yourself.

Proof. a) /by The Darboux Criterion/ $f \in (I) \Leftrightarrow (f \text{ is bounded on } I) \wedge (\underline{\mathcal{J}} = \overline{\mathcal{J}})$, where $\underline{\mathcal{J}} = \sup_P s(f, P)$, $\overline{\mathcal{J}} = \inf_P S(f, P)$. Notice that $\forall \varepsilon > 0, \exists P, \text{ s.t. } S(f, P) - s(f, P) < \varepsilon \Leftrightarrow \underline{\mathcal{J}} = \overline{\mathcal{J}}$ (proof is similar with 1-dimension condition), which is what we need to prove.

b) [Segmented estimation] Sufficiency: procedure is same with book, a) just simplify the last step.